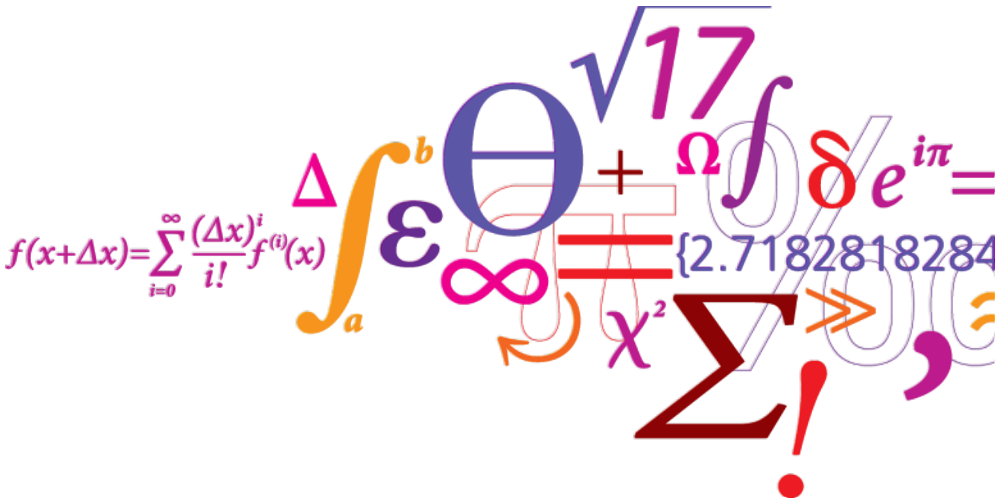
02526 Mathematical Modeling
Operations Research:
Modeling with integer variables

Stefan Røpke



DTU Management / DTU Compute

Agenda

- Optimization problems involving integer variables
- Julia – a programming language especially well suited for Operations Research
- The project

A bit about myself

- contact:
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  - Building 358, room 144
- Background
  - PhD from University of Copenhagen (Computer Science).
  - Studied for 2.5 years at University of Montreal (PhD/post doc).
  - Employed at DTU since 2008
  - Professor at DTU Management since 2012.
  - Research: Solution methods and modeling of complex optimization problems. Especially with applications within transport.

Linear programming
(”programming” = ”planning”/”optimization”)

- Known from 42101 Introduction to Operations Research

Linear programming - Example

- 2 products (glass doors, windows)
- Requires time on 3 workshops

	1	2	Capacity (hours/week)
1	1	0	4
2	0	2	12
3	3	2	18
profit	3	5	

- Profit is per unit and should be multiplied by 1000$
- We want to plan the production so that our profit is maximized

Linear programming - Example

Decision variables:
x_1 = production of product 1 (units/week)
x_2 = production af product 2 (units/week)

	1	2	Capacity (Hours/week)
1	1	0	4
2	0	2	12
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profit	3	5	

Linear programming - Example

Decision variables:
x_1 = production of product 1 (units/week)
x_2 = production af product 2 (units/week)

Symbolic model (LP)

$$\max Z = 3x_1 + 5x_2$$

subject to

$$\begin{aligned} x_1 &\leq 4 \\ 2x_2 &\leq 12 \\ 3x_1 + 2x_2 &\leq 18 \end{aligned}$$

and

$$x_1 \geq 0,\ x_2 \geq 0$$

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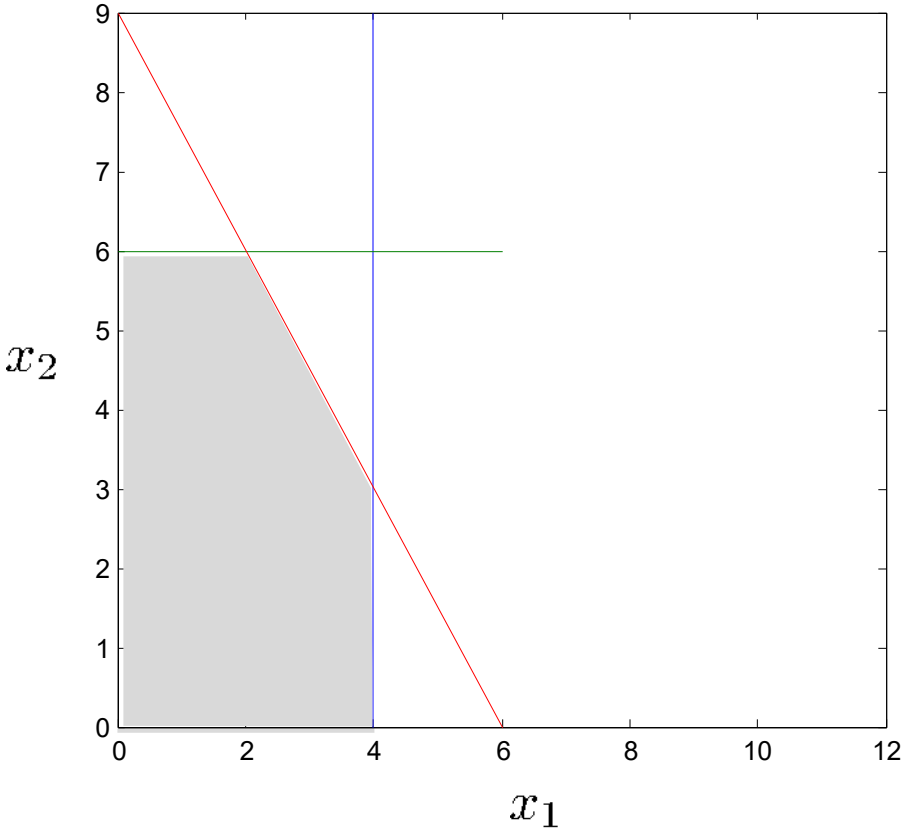
and

$$x_1 \geq 0,\ x_2 \geq 0$$

Optimal solution: $x_1 = 2, x_2 = 6, Z = 36$

	1	2	Capacity (Hours/week)
1	1	0	4
2	0	2	12
3	3	2	18
profit	3	5	

Linear programming - Example



$$\max Z = 3x_1 + 5x_2$$

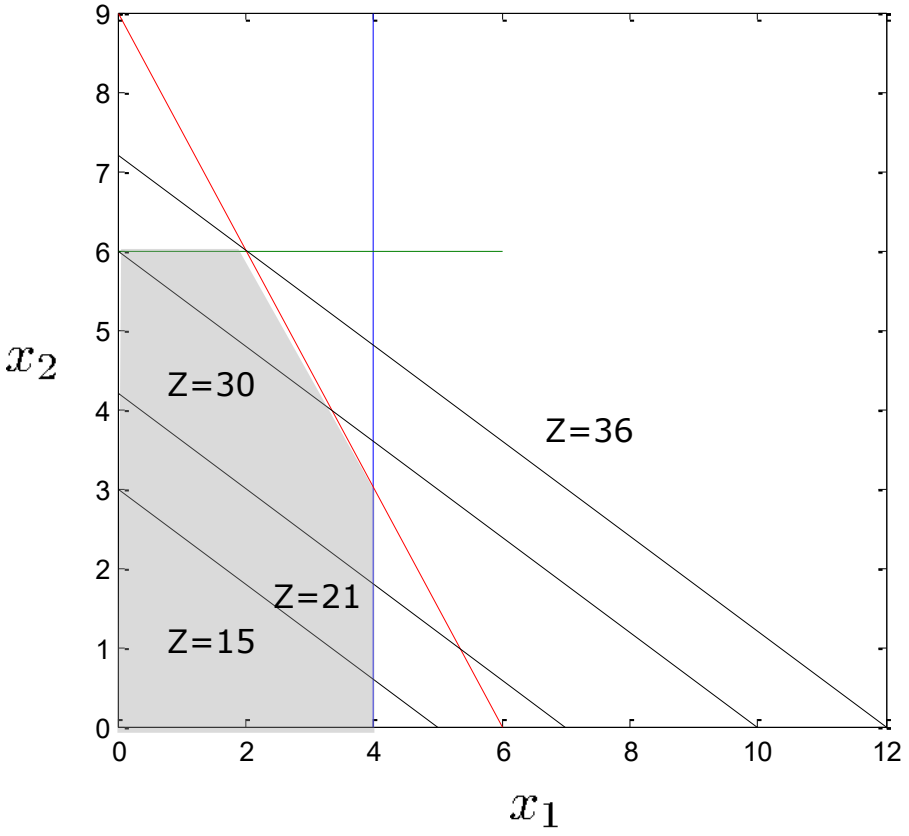
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subject to

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and

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Components of the model:

- Decision variables (what we are looking for)
- Objective function (what we wish to optimize)
- Constraints (that we have to respect)

	1	2	Capacity (Hours/week)
1	1	0	4
2	0	2	12
3	3	2	18
profit	3	5	

Linear programming (LP).

$$\max Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$$

subject to

$$\begin{aligned}
a_{11}x_1 &+a_{12}x_2 + \ldots +a_{1n}x_n \leq b_1\\
a_{21}x_1 &+a_{22}x_2 + \ldots +a_{2n}x_n \leq b_2\\
\vdots &\quad\vdots &\quad\vdots &\quad\vdots\\
a_{m1}x_1 &+a_{m2}x_2 + \ldots +a_{mn}x_n \leq b_m\\
x_1, &\quad x_2, \ldots, &\quad x_n \geq 0
\end{aligned}$$

c_j,a_{ij} and b_i are constants
x_j are variables

- We can also handle minimization, "=" and "$\geq$" constraints and negative variables.
- Linear problems are solved using the *Simplex Algorithm* and we are typically able to solve the problems that appear in practice.
- We **cannot** handle non-linearities and discrete variables. E.g.
  $x_1 \cdot x_2$
  x^2
  $|x_1 - x_2|$
  $x_4 \in \mathbb{Z}$
  are not allowed in linear programming.

Integer (linear) programming (IP)

- Like linear programming, but variables has to take integer values

$$\max Z = c_1 x_1 + c_2 x_2 + \ldots c_n x_n$$

subject to

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &\leq b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &\leq b_2 \\ \vdots \qquad & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &\leq b_m \\ x_1, x_2, \ldots, x_n &\in \mathbb{Z} \end{aligned}$$

More compactly written:

$$\max \sum_{j=1}^{n} c_j x_j$$

subject to

$$\begin{aligned} \sum_{j=1}^{n} a_{ij} x_j &\leq b_i \quad \forall i \in \{1, \ldots, m\} \\ x_j &\in \mathbb{Z} \quad \forall j \in \{1, \ldots, n\} \end{aligned}$$

Mixed integer (linear) programming (MIP)

- Like integer programming, but some variabels are forced to take integer variables while others can take real values

$$\max \sum_{j=1}^{n} c_j x_j$$

subject to

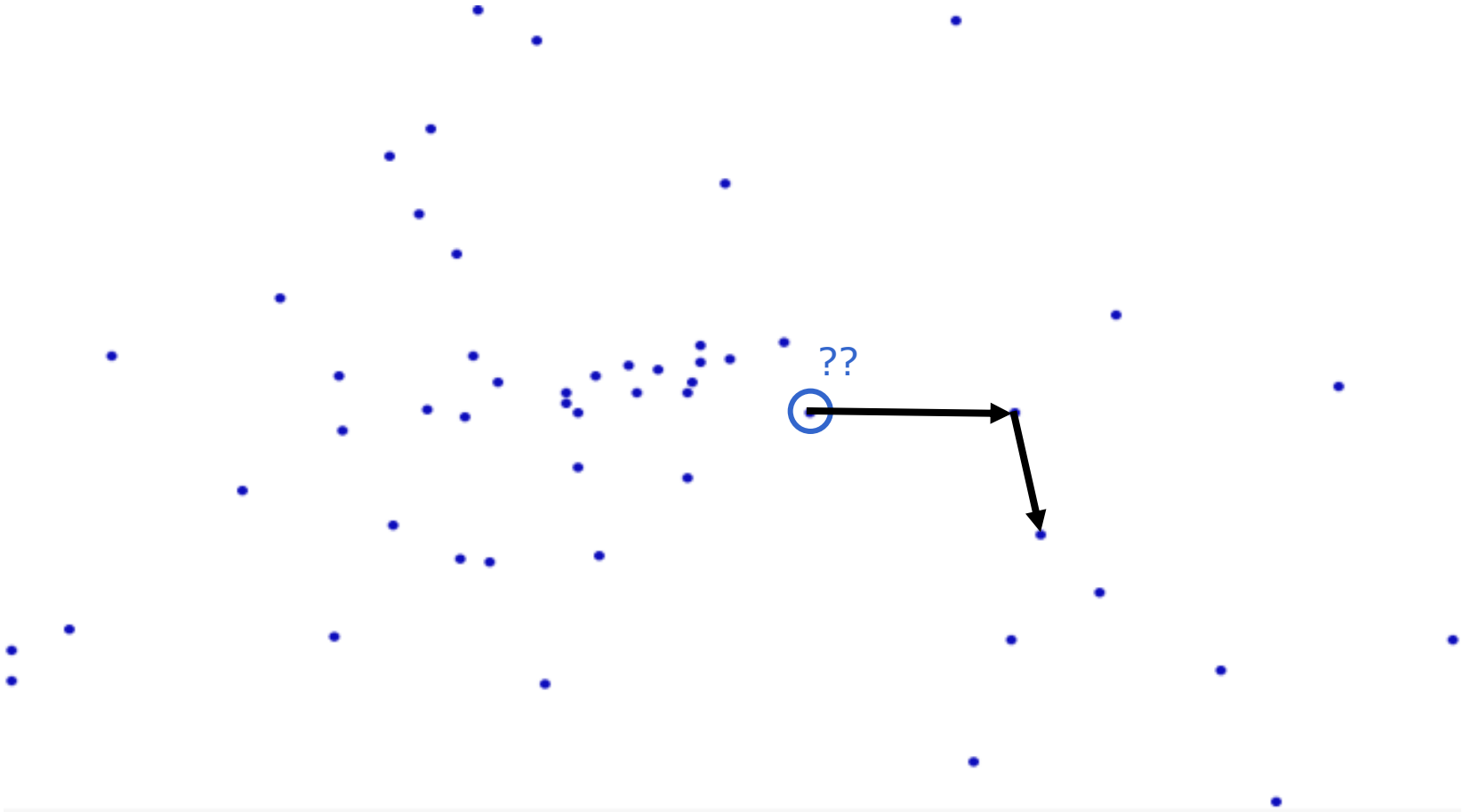
$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad \forall i \in \{1, \ldots, m\}$$

$$x_j \in \mathbb{Z} \quad \forall j \in S_1$$

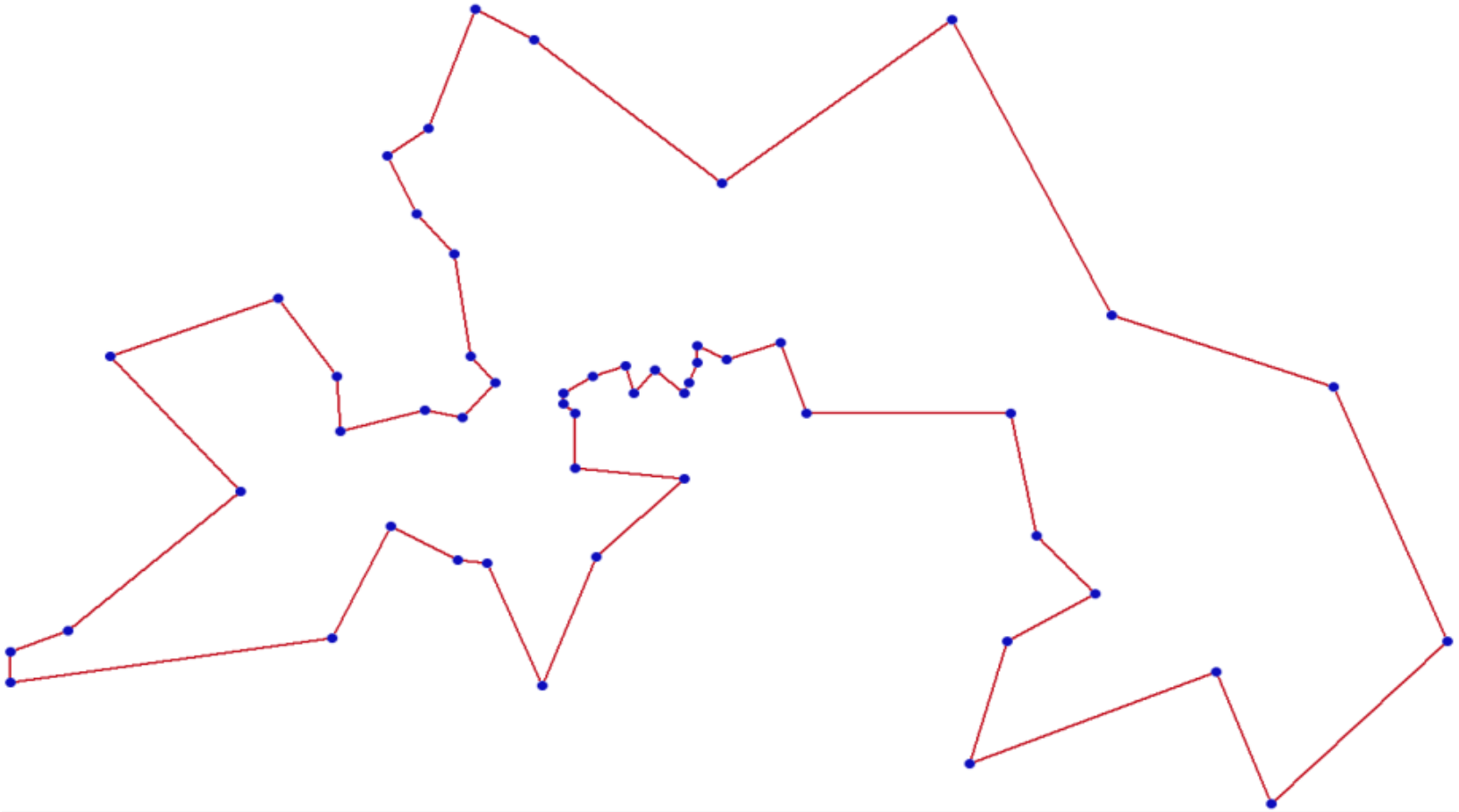
$$x_j \in \mathbb{R} \quad \forall j \in S_2$$

assumption: $S_1 \cap S_2 = \emptyset$

Applications:
The traveling salesman problem (TSP)



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The traveling salesman problem (TSP)

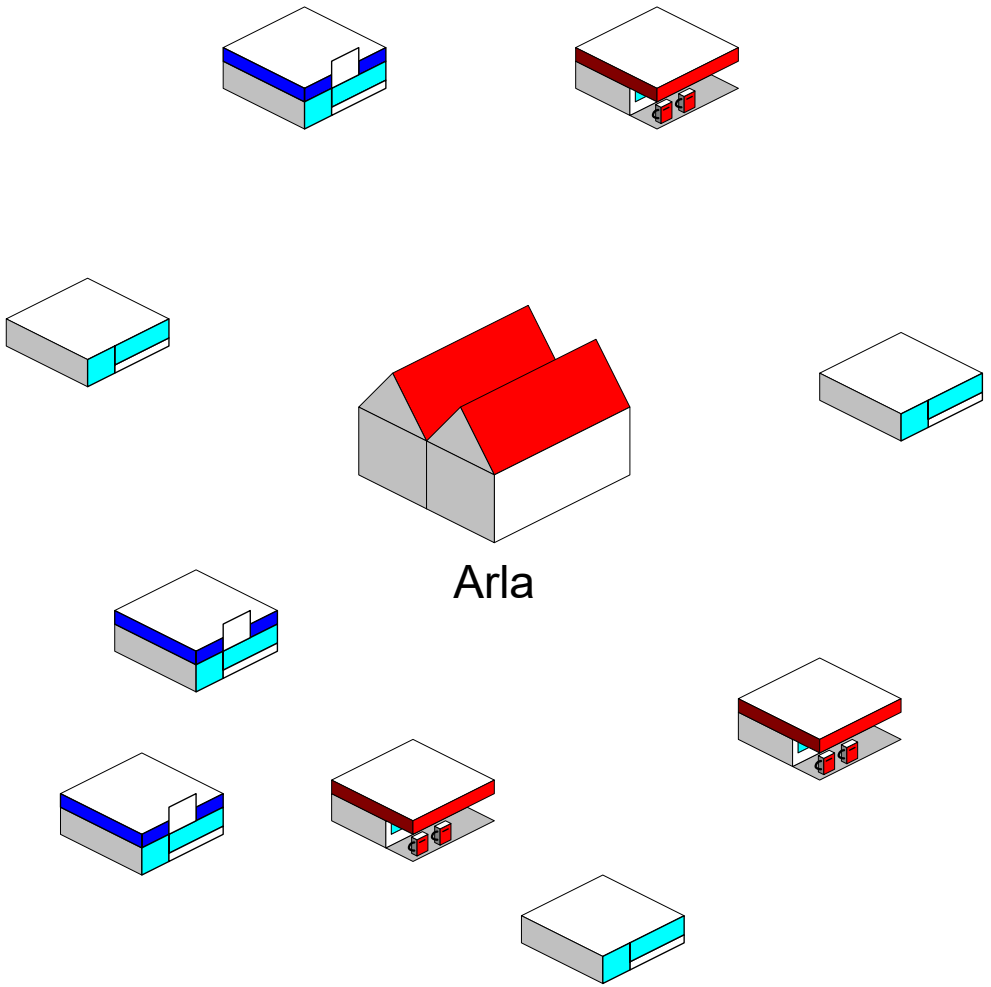


Applications : Vehicle routing

Our task: supply supermarkets, gas stations and corner stores with dairy products.

Assume at most 4 visits on each route

Construct routes that minimizes total distance traveled

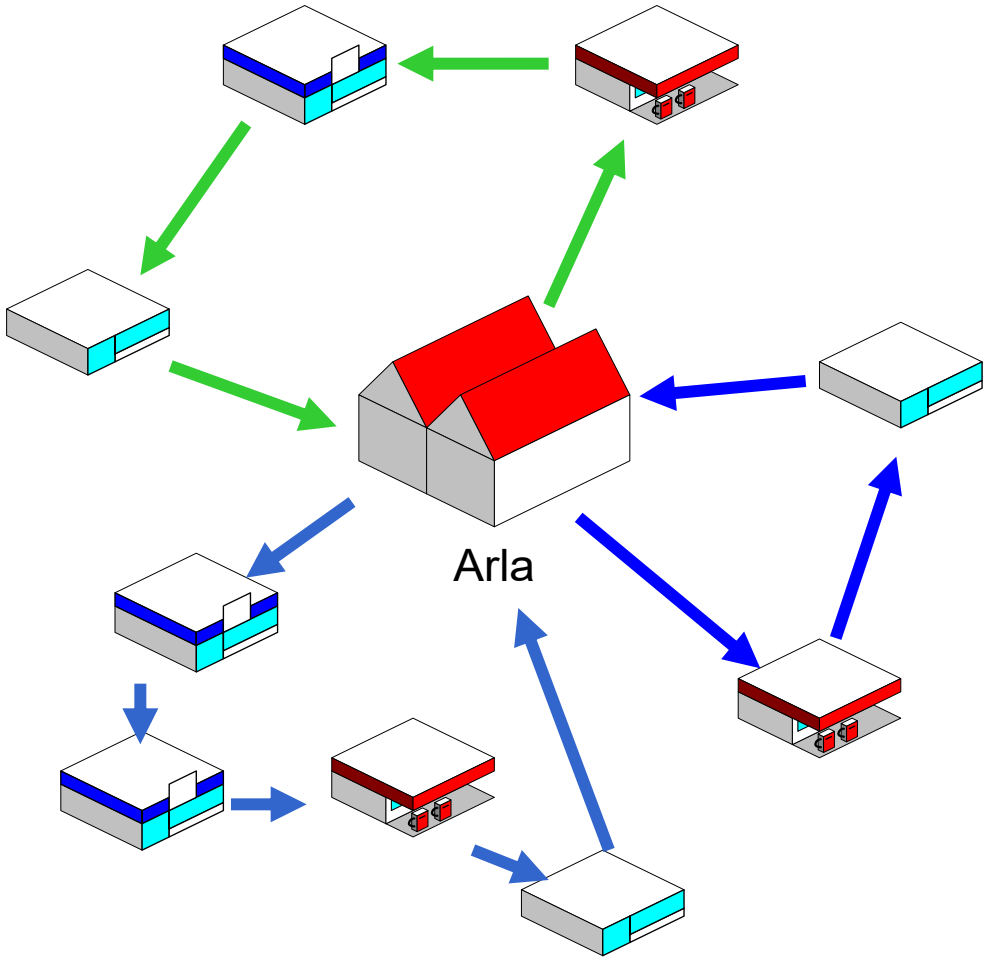


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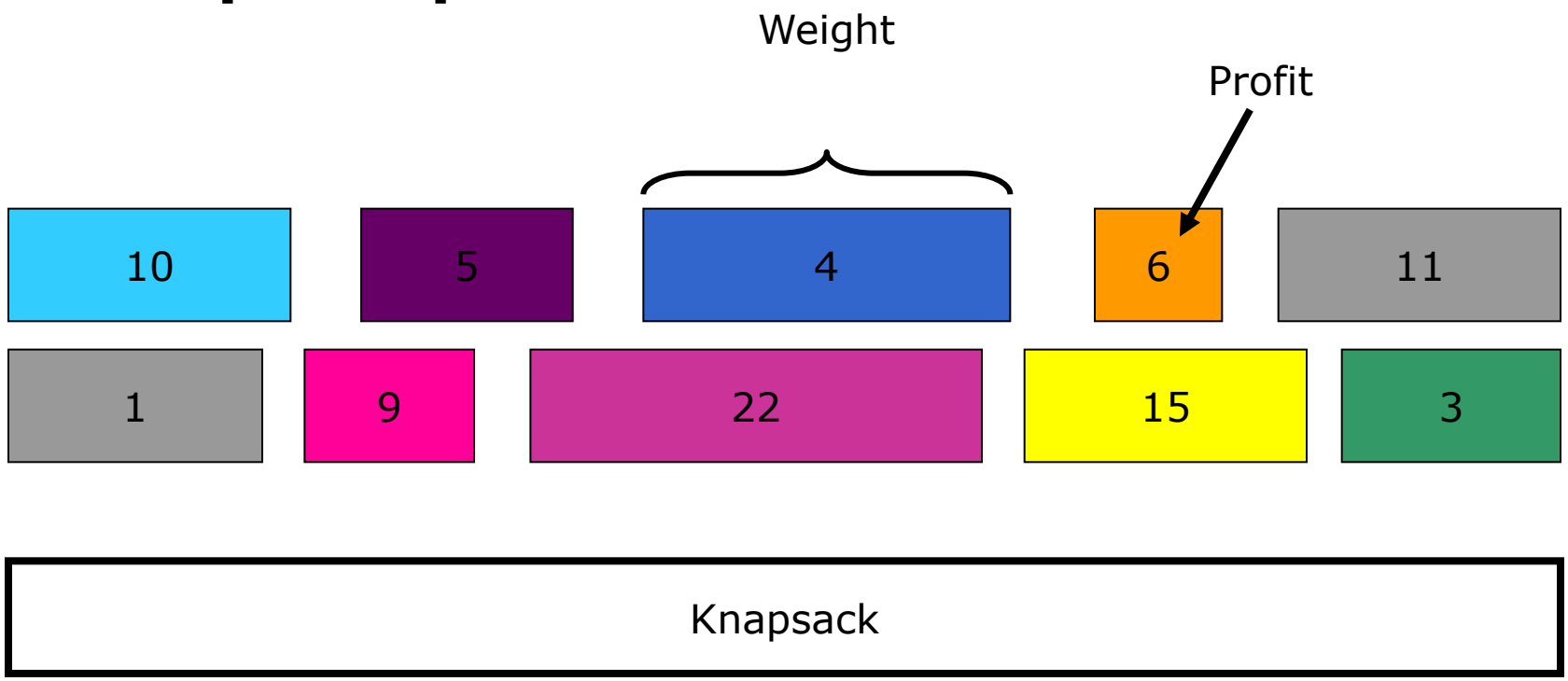
Construct routes that minimizes total distance traveled



Integer programming and Operations Research (OR) can be found in your everyday life.

- Many products your buy have been transported using trucks and ships. Most likely OR has been used for planning part of the transport.
- OR could have been used for planning the production of the products – or planning the work shifts at the production sites.
- When you take the bus or the train OR could have been used for planning the time table.
- Layout of windfarms and routing of cables between turbines are done using OR techniques.

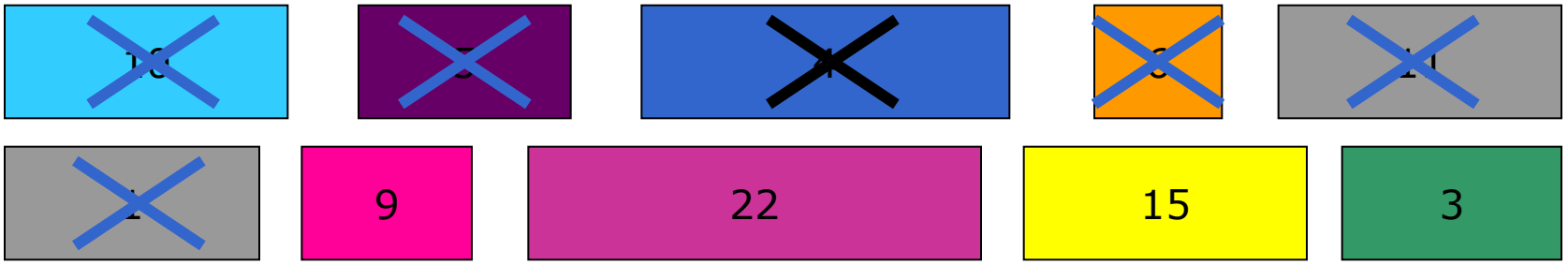
Some simple integer programs: The knapsack problem



Goal: maximize profit of the chosen items. The weight of the selected items cannot exceed the capacity of the knapsack.

Knapsack problem

A feasible solution, pick items in the order they appear:

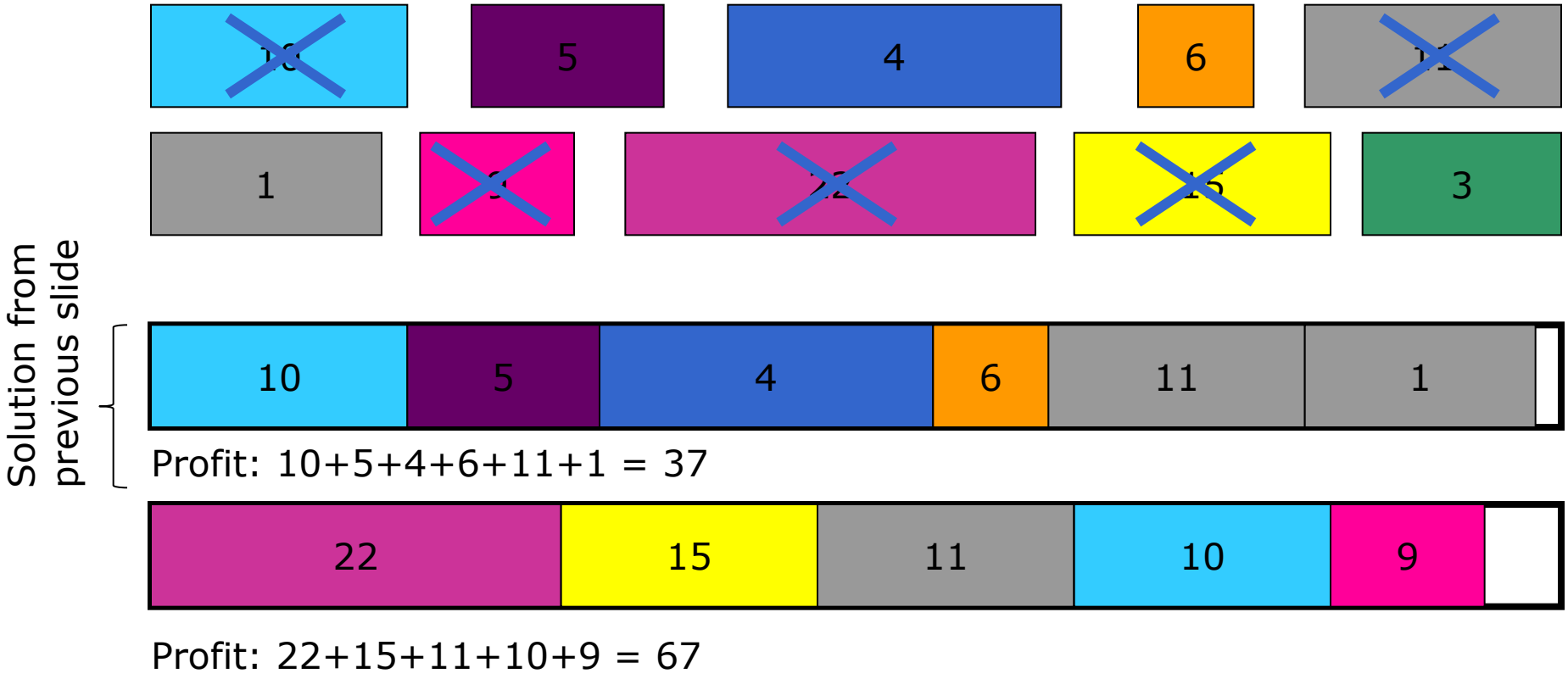




Profit: 10+5+4+6+11+1 = 37

Knapsack problem

A better solution: pick items in order of decreasing profit:



Some simple integer programs: The knapsack problem

Parameters and sets:

n : number of items.

$N = \{1, \ldots, n\}$: set of all items.

p_i : profit of selecting item i ($i \in N$).

w_i : weight of item i ($i \in N$).

c : knapsack capacity.

Decision variables:

x_i : 1 if item i is added to the knapsack, 0 otherwise.

The model:

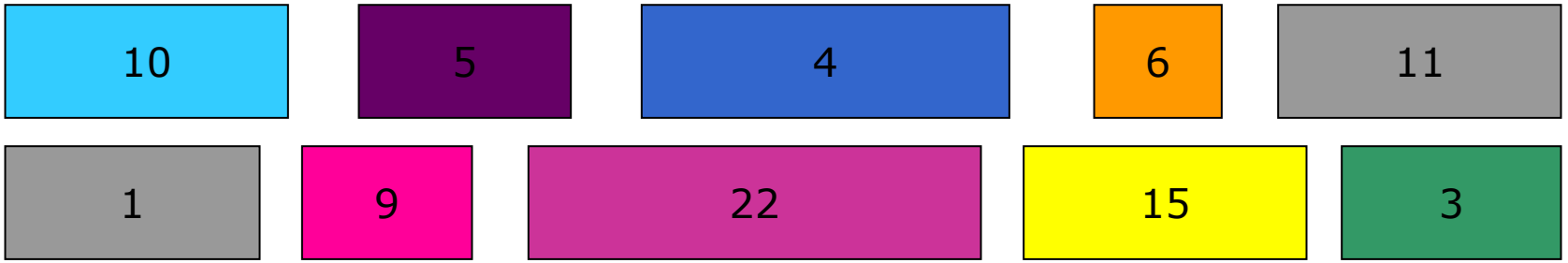
$$\max \sum_{i \in N} p_i x_i$$

subject to

$$\sum_{i \in N} w_i x_i \leq c$$

$$x_i \in \{0, 1\} \quad \forall i \in N$$

Example



Knapsack

Item	1	2	3	4	5	6	7	8	9	10
Profit	10	5	4	6	11	1	9	22	15	3
Weight	40	30	52	18	40	36	24	64	40	31

Capacity: 220

$$\max 10x_1 + 5x_2 + 4x_3 + 6x_4 + 11x_5 + 1x_6 + 9x_7 + 22x_8 + 15x_9 + 3x_{10}$$

subject to

$$40x_1 + 30x_2 + 52x_3 + 18x_4 + 40x_5 + 36x_6 + 24x_7 + 64x_8 + 40x_9 + 31x_{10} \leq 220$$

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10} \in \{0, 1\}$$

Example

Earlier solutions:

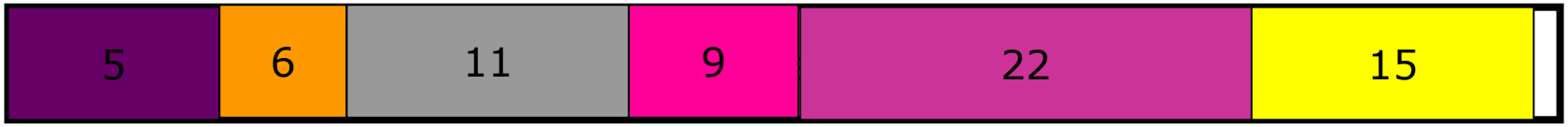


Profit: 10+5+4+6+11+1 = 37



Profit: 22+15+11+10+9 = 67

Optimal solution:



Profit: 68

Some simple integer programs: Capacitated facility location

Parameters	
f_i	cost of opening facility $i \in I$
c_{ij}	cost of serving customer $j \in J$ using facility $i \in I$
Q_i	capacity of facility $i \in I$.
d_j	demand of customer $j \in J$
Decision variables	
$y_i \in \{0,1\}$	1 if facility $i \in I$ is opened
$x_{ij} \in \{0,1\}$	1 if customer j is served by facility i. 0 otherwise.

$$\min \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} + \sum_{i \in I} f_i y_i$$

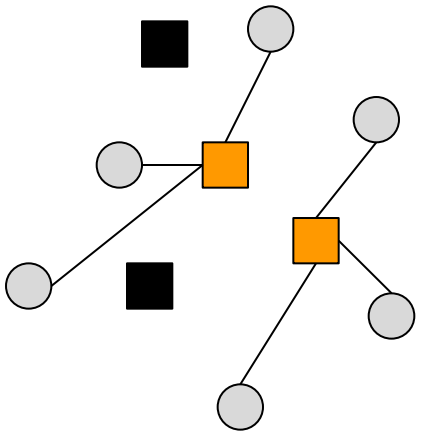
subject to

$$\sum_{i \in I} x_{ij} = 1 \qquad \forall j \in J$$

$$\sum_{j \in J} d_j x_{ij} \leq Q_i y_i \qquad \forall i \in I$$

$$x_{ij} \in \{0,1\} \qquad \forall i \in I, j \in J$$

$$y_i \in \{0,1\} \qquad \forall i \in I$$



Connection between LP and IP

Integer programming

$$Z_{IP} = \min \sum_{j=1}^{n} c_j x_j$$

subject to

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad \forall i \in \{1, \ldots, m\}$$

$$x_j \in \mathbb{Z}_+ \quad \forall j \in \{1, \ldots, n\}$$

Linear programming

$$Z_{LP} = \min \sum_{j=1}^{n} c_j x_j$$

subject to

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad \forall i \in \{1, \ldots, m\}$$

$$x_j \in \mathbb{R}_+ \quad \forall j \in \{1, \ldots, n\}$$

What can you say about Z_{LP} and Z_{IP}?

- $Z_{LP} = Z_{IP}$?
- $Z_{LP} \leq Z_{IP}$?
- $Z_{LP} \geq Z_{IP}$?
- You cannot say anything in general about Z_{LP} and Z_{IP}

Connection between LP and IP

Integer programming

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What can you say about Z_{LP} and Z_{IP}?

- $Z_{LP} = Z_{IP}$?
- $Z_{LP} \leq Z_{IP}$?  True answer
- $Z_{LP} \geq Z_{IP}$?
- You cannot say anything in general about Z_{LP} and Z_{IP}

Connetion between LP and IP

LP

$$\min -2x_1 + 3x_2$$

subject to

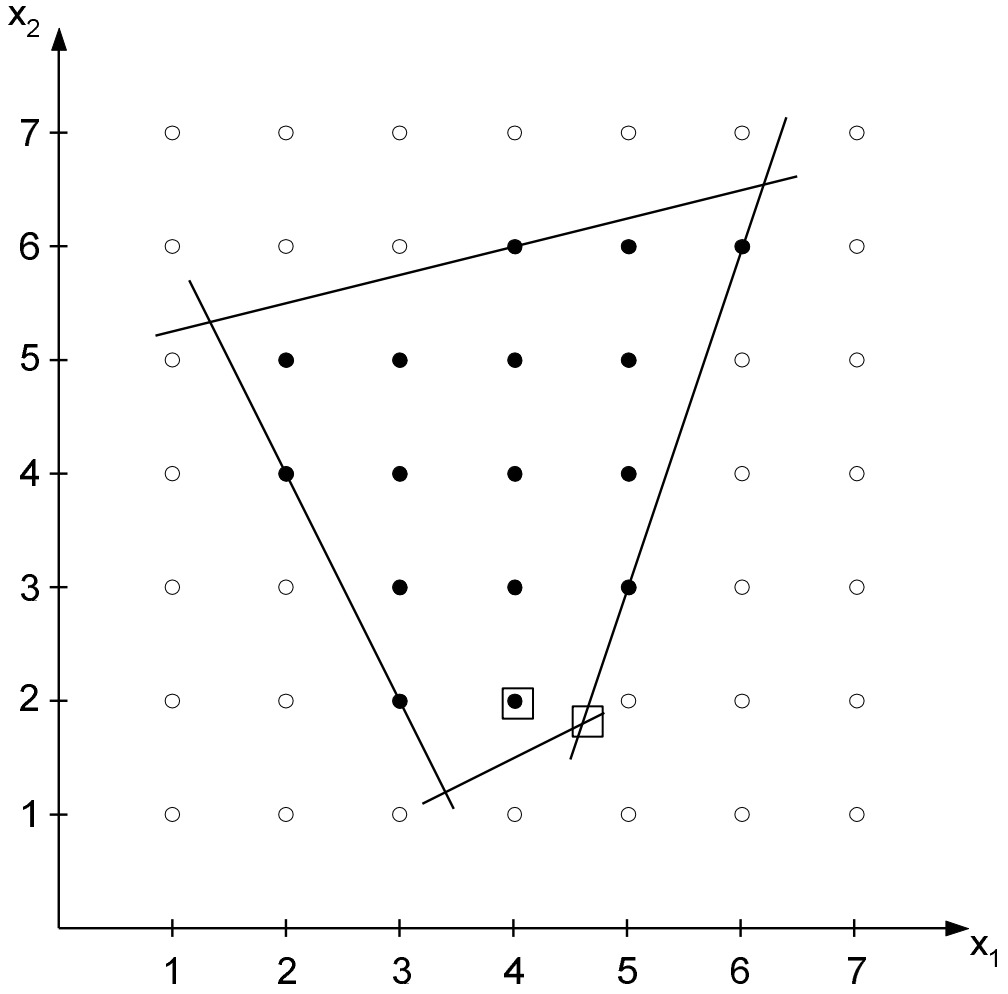
$$\begin{aligned} 2x_1 + x_2 &\geq 8 && (1)\\ -3x_1 + x_2 &\geq -12 && (2)\\ x_1 - 4x_2 &\geq -20 && (3)\\ -x_1 + 2x_2 &\geq -1 && (4)\\ x_1, x_2 &\in \mathbb{R} && (5) \end{aligned}$$

IP

$$\min -2x_1 + 3x_2$$

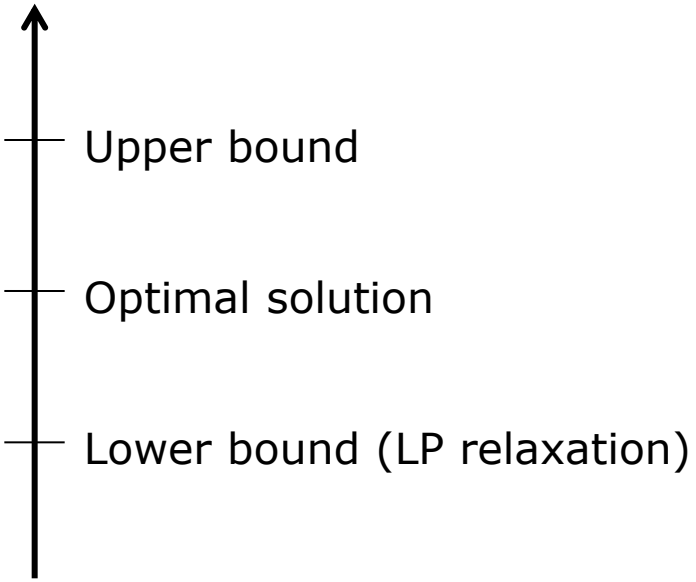
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Branch and bound

- Assume: Solve minimizing IP.
- Lower bound: Solve LP version of IP (”LP relaxation”)
- Upper bound: Any feasible solution to the IP if we can find such a solution



Solve capacitated facility location using branch and bound
3 warehouses, 20 customers, f_i=2000, c_{ij}= Euclidean distance

Parameters	
f_i	cost of opening facility $i \in I$
c_{ij}	cost of serving customer $j \in J$ using facility $i \in I$
Q_i	capacity of facility $i \in I$.
d_j	demand of customer $j \in J$
Decision variables	
$y_i \in \{0,1\}$	1 if facility $i \in I$ is opened
$x_{ij} \in \{0,1\}$	1 if customer j is served by facility i. 0 otherwise.

$$\min \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} + \sum_{i \in I} f_i y_i$$

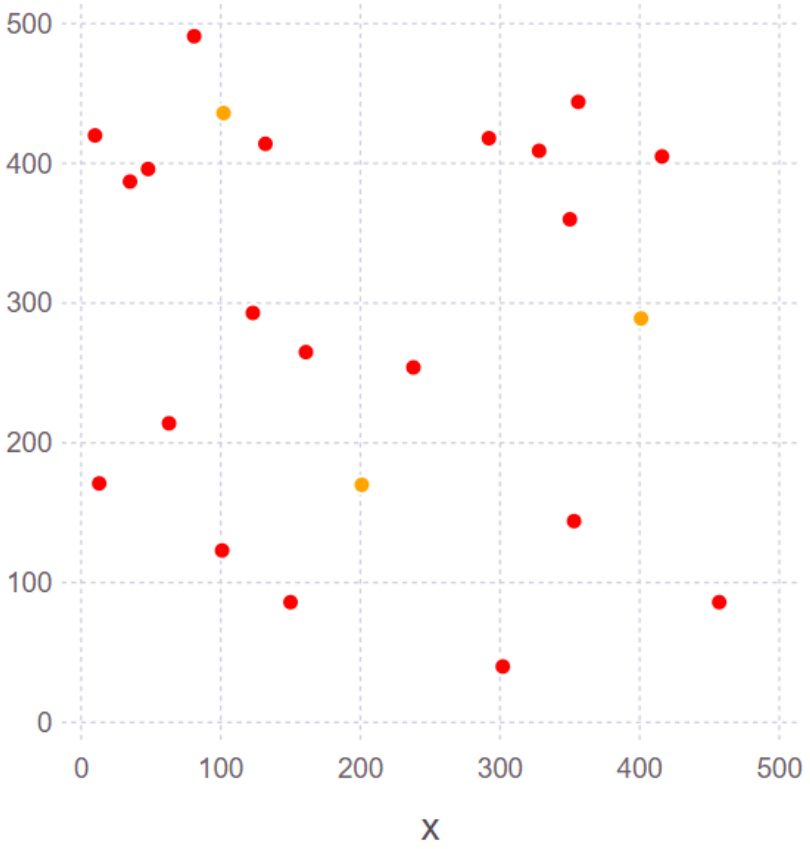
subject to

$$\sum_{i \in I} x_{ij} = 1 \qquad \forall j \in J$$

$$\sum_{j \in J} d_j x_{ij} \leq Q y_i \qquad \forall i \in I$$

$$x_{ij} \in \{0,1\} \quad \forall i \in I, j \in J$$

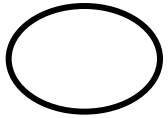
$$y_i \in \{0,1\} \quad \forall i \in I$$



i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	$Q_i = 191$
d_i	17	10	16	17	17	14	18	10	11	11	14	15	13	17	11	11	20	10	17	18	

Branch and bound

Solve LP relaxation -> Lower bound (LB)
Find feasible solution to IP -> Upper bound (UB)



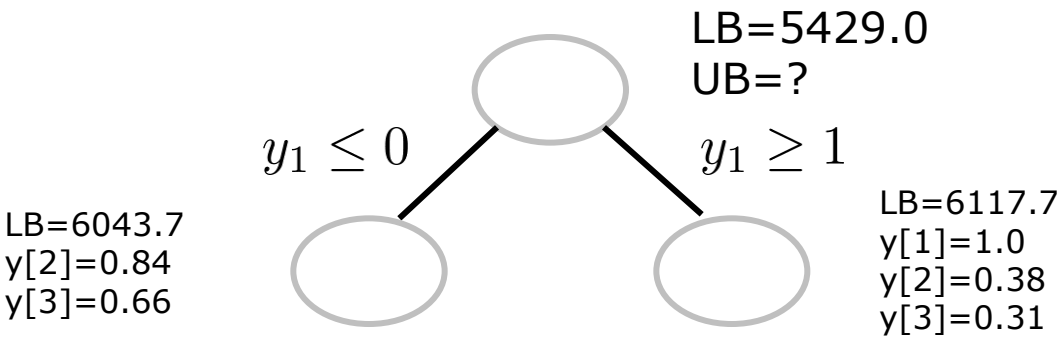
LB=5429.0
UB=?

y[1]=0.52
y[2]=0.58
y[3]=0.40

- We cannot decide which solution is optimal based on the available information.
- Split solution space in two:
- Find a variable that takes a non-integer value in the LP solution. Let's use y_1.
- One sub-problem is constructed by adding $y_1 \leq 0$ to the set of constraints. The other sub-problem is constructed by adding $y_1 \geq 1$ to the set of constraints.

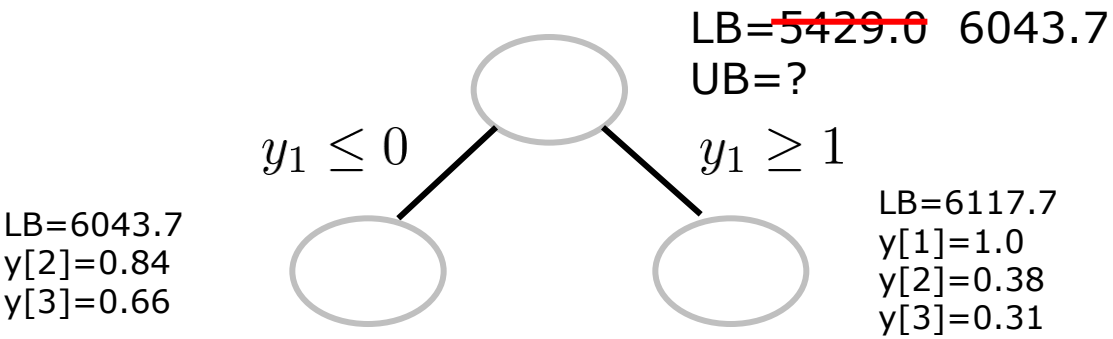
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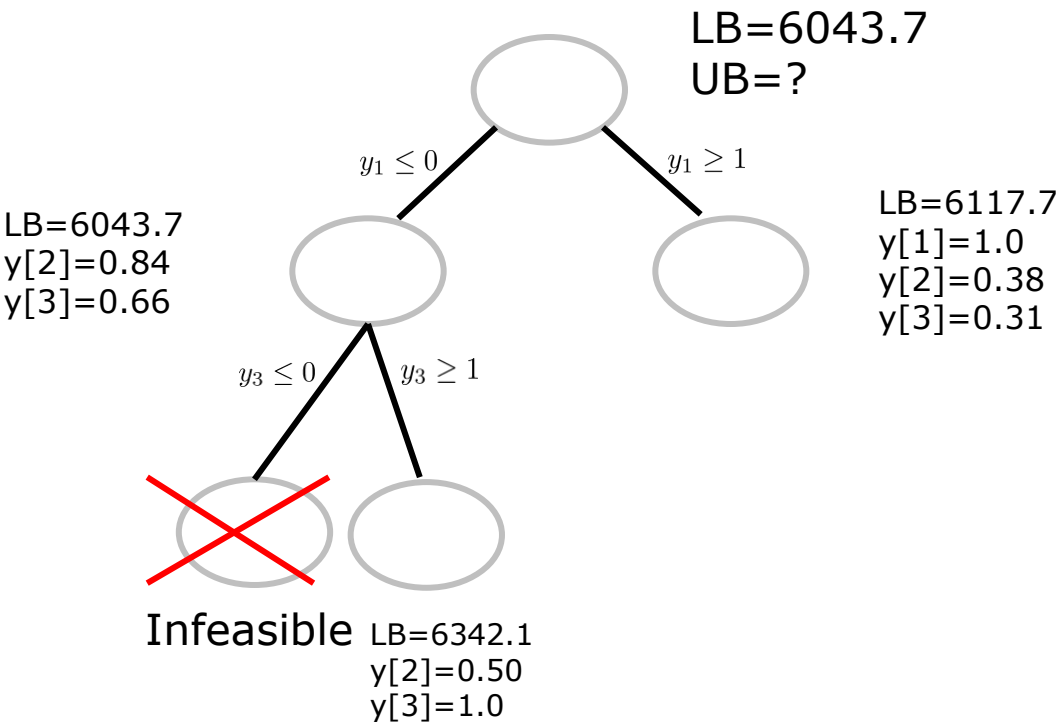
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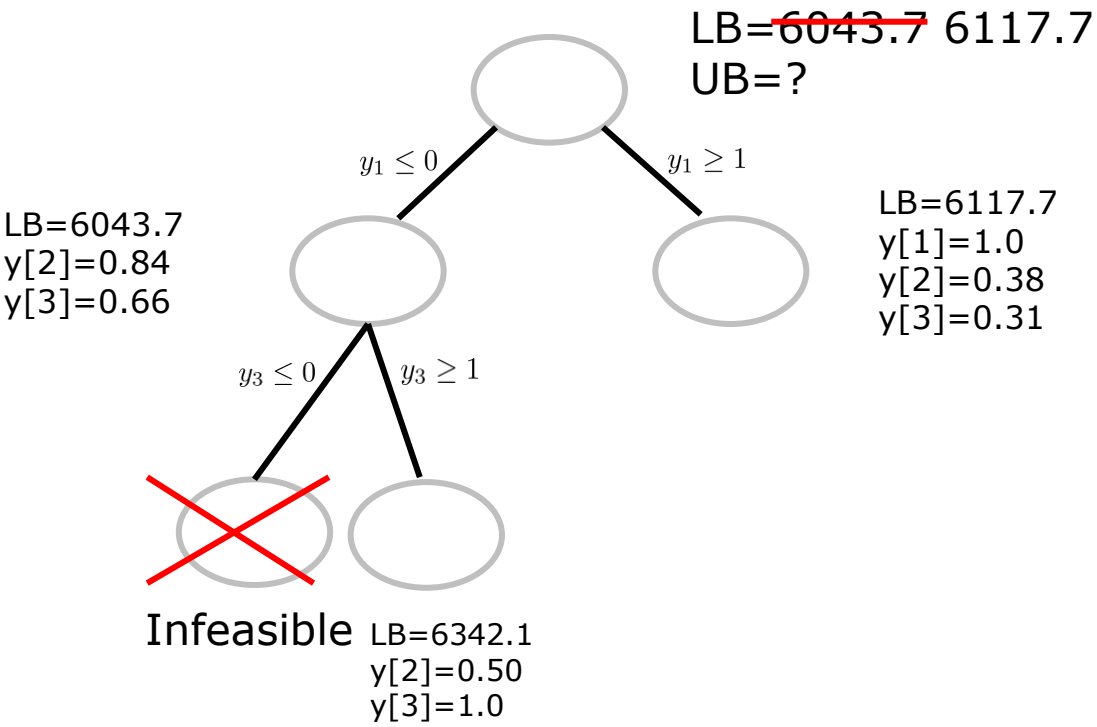
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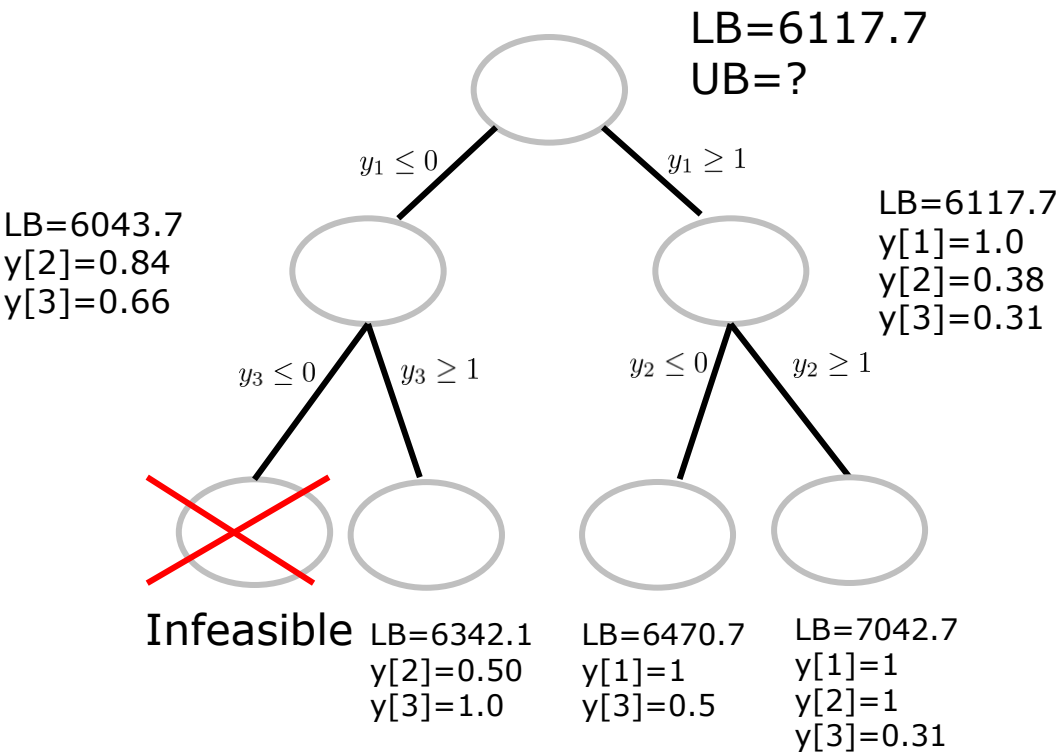
Branch and bound

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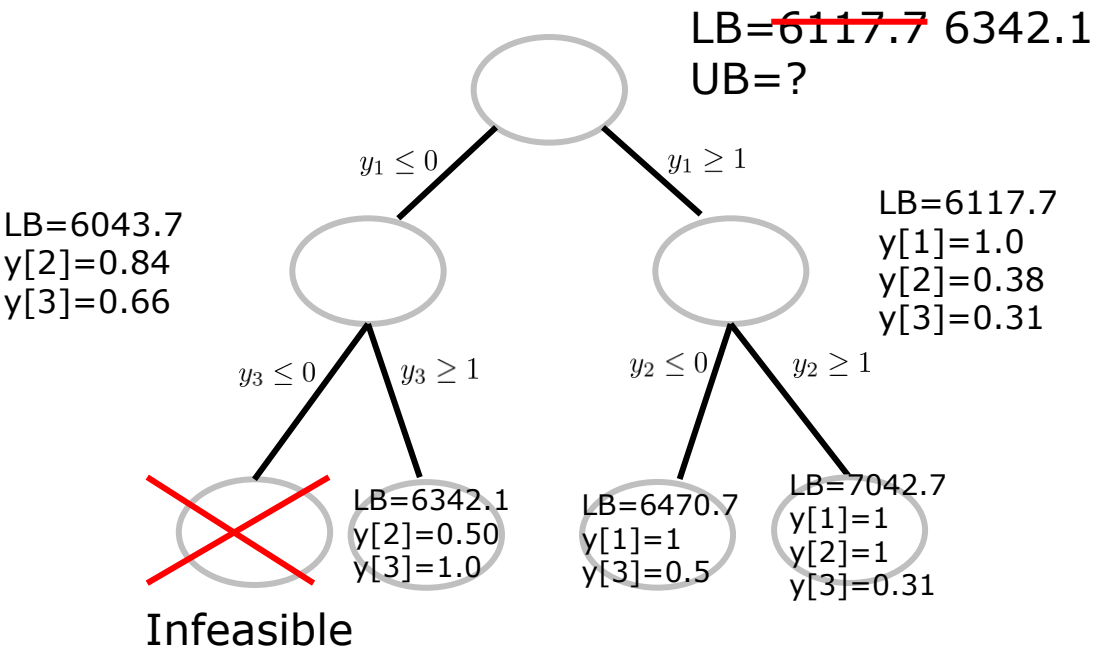
Branch and bound

Solve LP relaxation -> Lower bound (LB)
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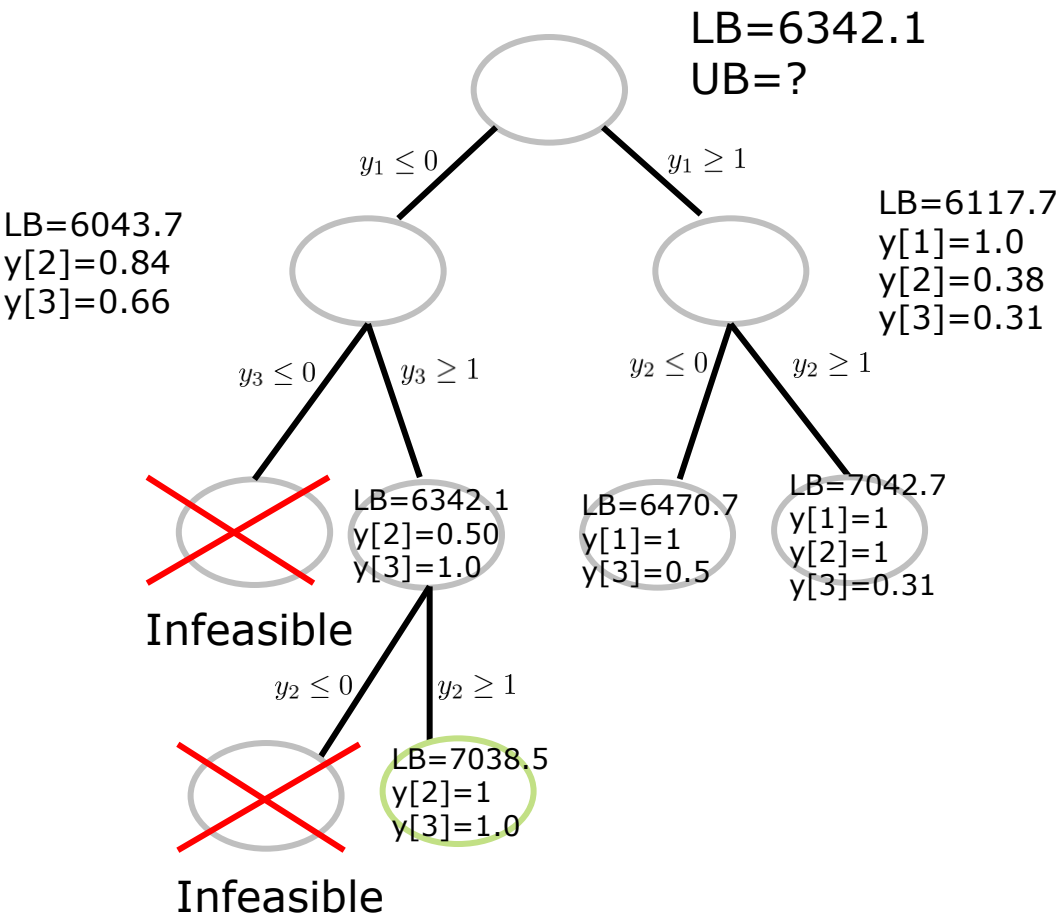
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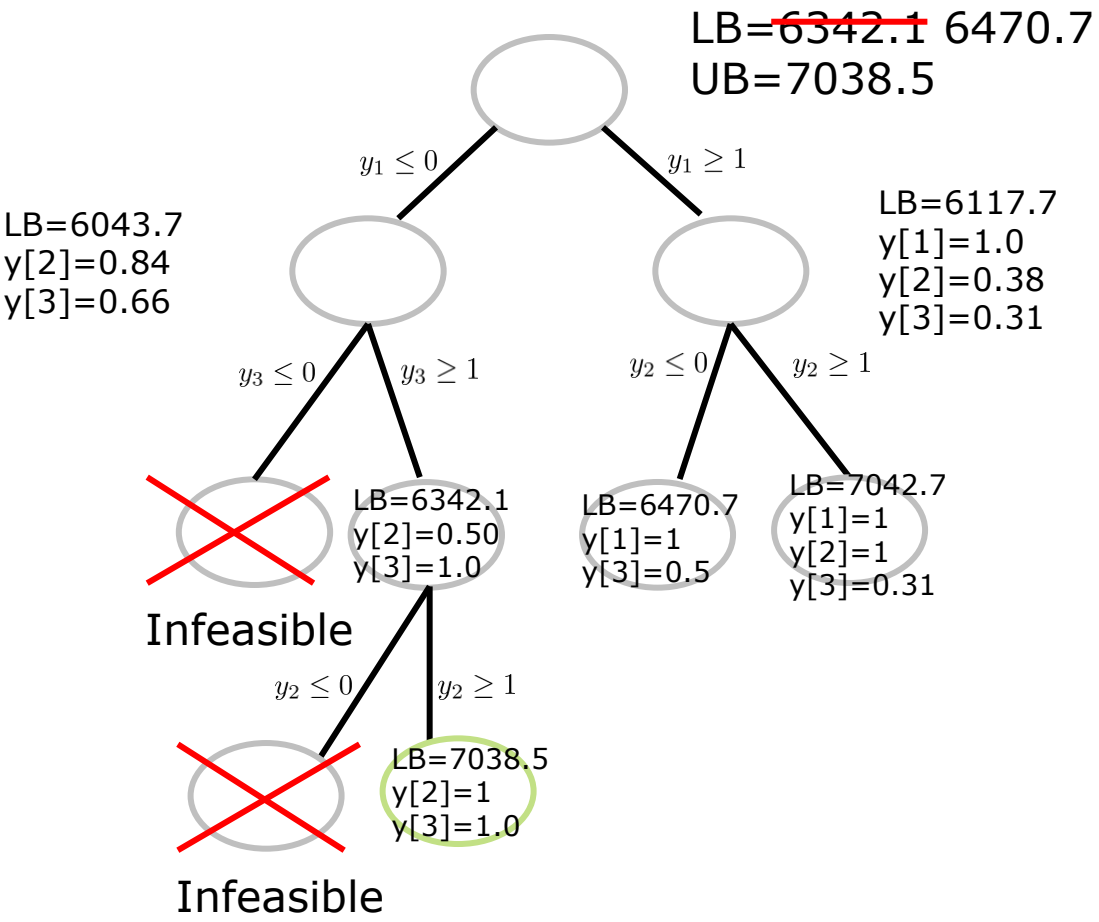
Branch and bound

Solve LP relaxation -> Lower bound (LB)
Find feasible solution to IP -> Upper bound (UB)



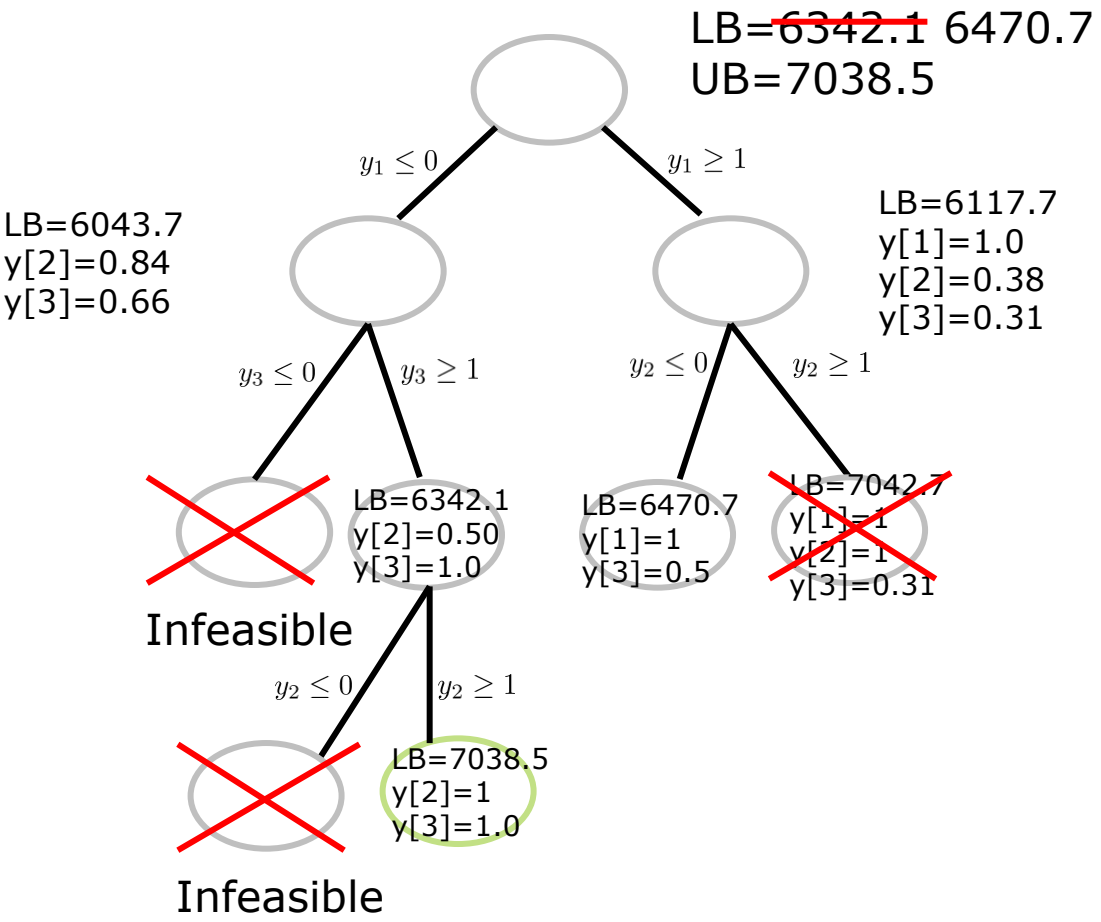
Branch and bound

Solve LP relaxation -> Lower bound (LB)
Find feasible solution to IP -> Upper bound (UB)



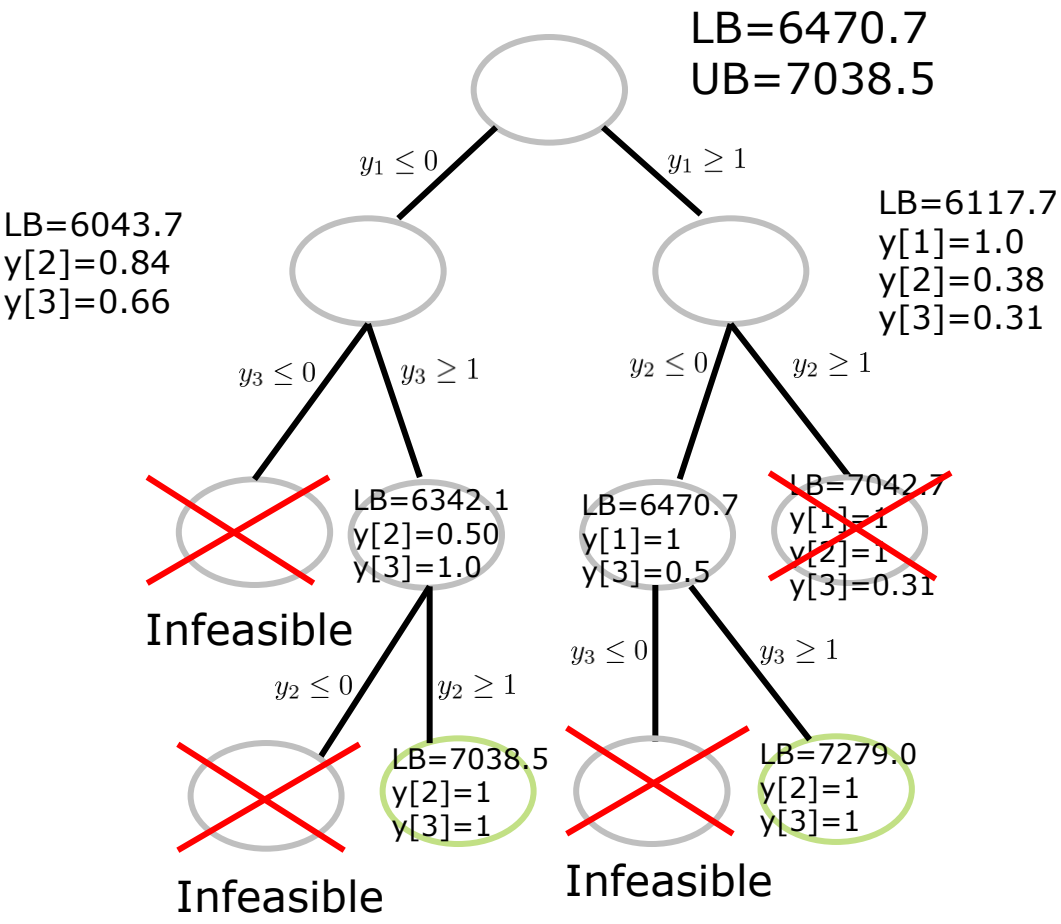
Branch and bound

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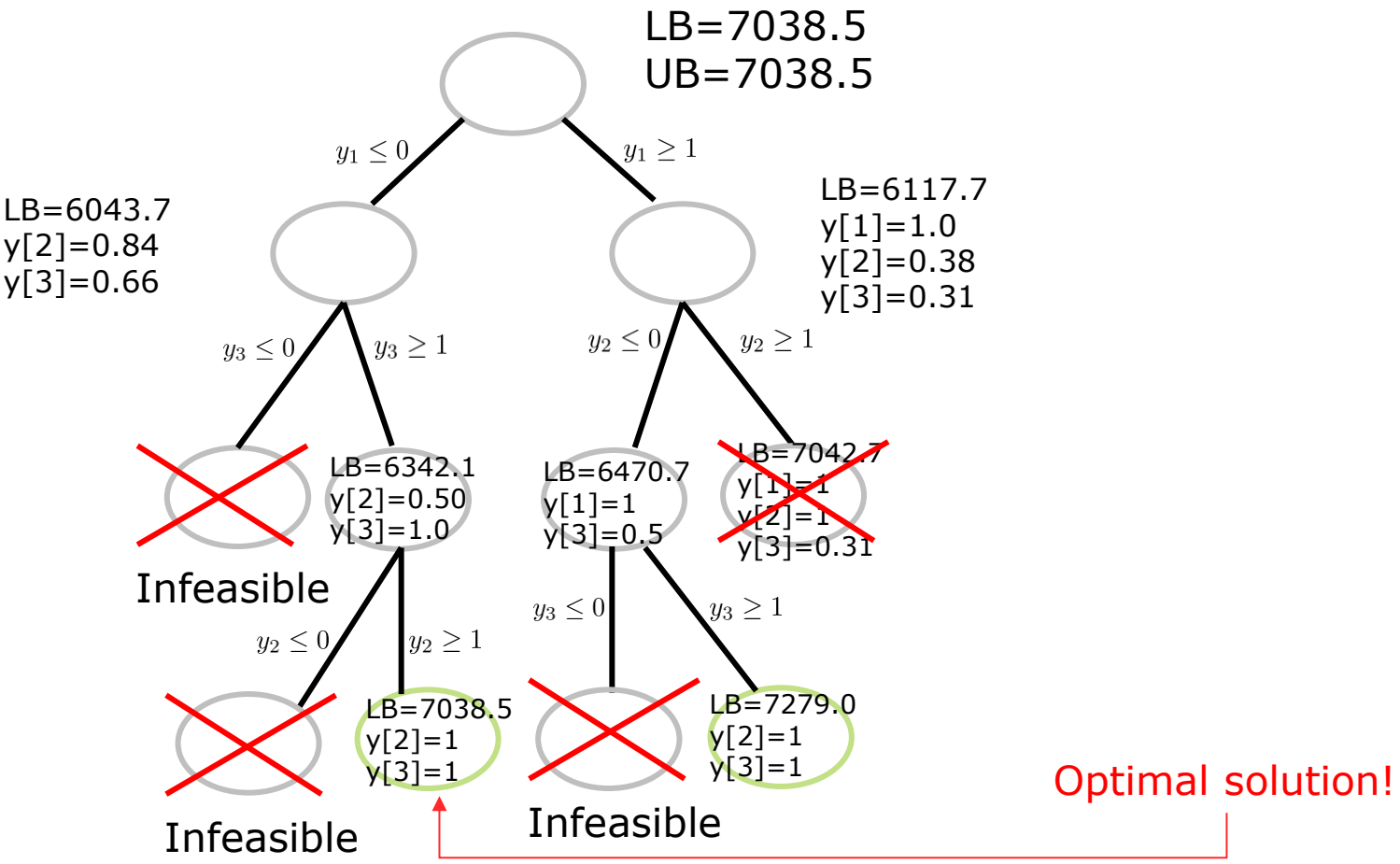
Branch and bound

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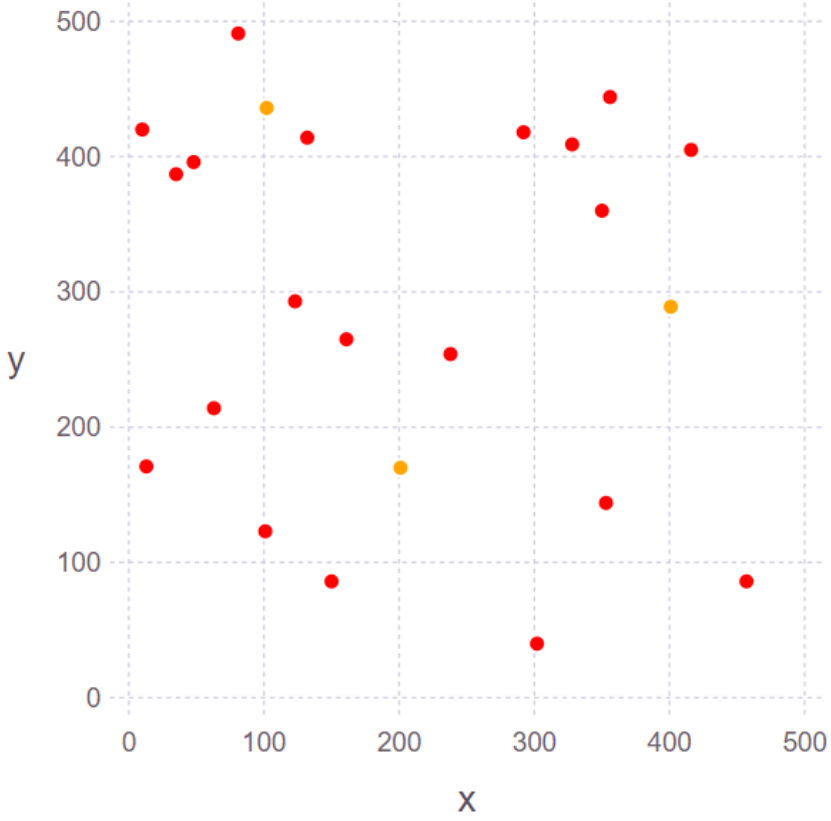


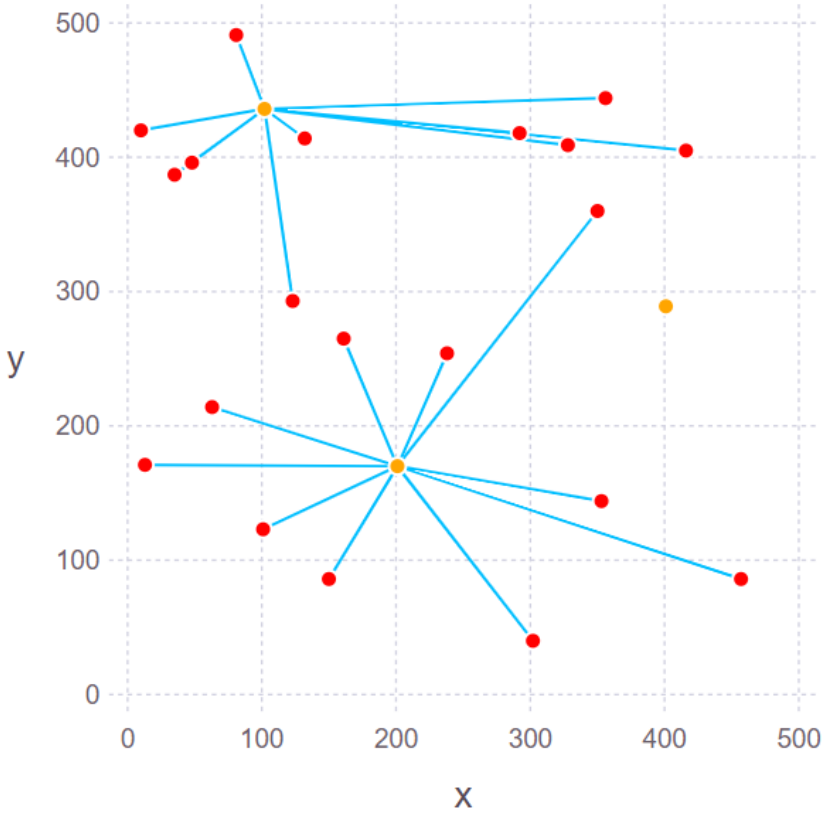
Branch and bound

Solve LP relaxation -> Lower bound (LB)
Find feasible solution to IP -> Upper bound (UB)



Optimal solution
3 warehouses, 20 customers, f_i=2000, c_{ij}= Euclidean distance





More about Operations research and integer programming

- **42101 Operations Research (Fall)**
  - Linear programming, Introduction to integer programming, theory and methods
- **42114 Integer programming (fall)**
  - Theory and methods
- **42112 Mathematical Programming Modelling (january)**
  - Modeling